Retarded Potentials and Far-Field Radiation

Source: D. Tong, Electromagnetism, Problem Sheet 7, Question 4.

Let ϕ be the retarded scalar potential given by

$$\phi(t,\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\int d^3y\,\frac{\rho(t_{\rm ret},\mathbf{y})}{R}, \tag{1.1}$$

where $R = |\mathbf{x}-\mathbf{y}|$, $t_{\rm ret} = t - R/c$, and $\hat{\mathbf{R}} = (\mathbf{x}-\mathbf{y})/R$. Show that

$$\frac{\partial}{\partial t}\phi(t,\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\int d^3y\,\frac{\dot\rho(t_{\rm ret},\mathbf{y})}{R}, \tag{1.2}$$

where $\dot\rho(t_{\rm ret},\mathbf{y})$ is $\partial\rho(t,\mathbf{y})/\partial t$ evaluated at $t_{\rm ret}$. Show further that

$$\nabla\phi(t,\mathbf{x}) = -\frac{1}{4\pi\varepsilon_0}\int d^3y\,\hat{\mathbf{R}}\left(\frac{1}{R^2}\rho(t_{\rm ret},\mathbf{y}) + \frac{1}{cR}\dot\rho(t_{\rm ret},\mathbf{y})\right). \tag{1.3}$$

Hence verify, using $\nabla^2(1/R) = -4\pi\delta^{(3)}(\mathbf{x}-\mathbf{y})$, that ϕ satisfies the wave equation:

$$\Box\,\phi(t,\mathbf{x}) = -\frac{1}{\varepsilon_0}\rho(t,\mathbf{x}). \tag{1.4}$$

Write down a similar retarded solution for the vector potential $\mathbf{A}$ in terms of the current density $\mathbf{J}$.

Now assume that ρ and $\mathbf{J}$ are non-zero only in a finite region. Setting $\hat{\mathbf{x}} = \mathbf{x}/|\mathbf{x}|$, show that the leading terms in the far-field expansion are

$$\begin{aligned}\mathbf{E}(t,\mathbf{x}) &\approx \frac{\mu_0}{4\pi|\mathbf{x}|}\int d^3y\left(\hat{\mathbf{x}}c\dot\rho(t_{\rm ret},\mathbf{y}) - \dot{\mathbf{J}}(t_{\rm ret},\mathbf{y})\right)\\ &= \frac{\mu_0}{4\pi|\mathbf{x}|}\,\hat{\mathbf{x}}\times\left(\hat{\mathbf{x}}\times\int d^3y\,\dot{\mathbf{J}}(t_{\rm ret},\mathbf{y})\right),\end{aligned} \tag{1.5}$$

where conservation of current, integration by parts, and discarding of a surface integral have all been used, and

$$\mathbf{B}(t,\mathbf{x}) \approx -\frac{\mu_0}{4\pi c|\mathbf{x}|}\,\hat{\mathbf{x}}\times\int d^3y\,\dot{\mathbf{J}}(t_{\rm ret},\mathbf{y}) = \frac{1}{c}\hat{\mathbf{x}}\times\mathbf{E}(t,\mathbf{x}). \tag{1.6}$$

(Note that these results do not assume the dipole approximation.) Determine the Poynting vector.

Hint: when using current conservation, and integration by parts, be careful with the $\mathbf{y}$-dependence of $t_{\rm ret}$.

Solution. Following from the retarded potential, we have, trivially,

$$\frac{\partial}{\partial t}\phi(t,\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\int d^3y\,\frac{\dot{\rho}(t_{\text{ret}},\mathbf{y})}{R}. \tag{1.7}$$

Then, note that

$$\nabla\left(\frac{1}{R}\right) = -\frac{\hat{\mathbf{R}}}{R^2} \qquad \text{and} \qquad \nabla\rho(t_{\text{ret}},\mathbf{y}) = \frac{\partial\rho(t_{\text{ret}},\mathbf{y})}{\partial R} = -\frac{1}{c}\dot{\rho}(t_{\text{ret}},\mathbf{y})\hat{\mathbf{R}}. \tag{1.8}$$

Thus, we have

$$\nabla\phi(t,\mathbf{x}) = -\frac{1}{4\pi\varepsilon_0}\int d^3y\,\hat{\mathbf{R}}\left(\frac{1}{R^2}\rho(t_{\text{ret}},\mathbf{y}) + \frac{1}{cR}\dot{\rho}(t_{\text{ret}},\mathbf{y})\right). \tag{1.9}$$

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Now, first consider the second-order time derivative, which is simply given by

$$\frac{\partial^2}{\partial t^2}\phi(t,\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\int d^3y\,\frac{\ddot{\rho}(t_{\text{ret}},\mathbf{y})}{R}. \tag{1.10}$$

For the Laplacian, it is slightly trickier; we find

$$\nabla^2\phi(t,\mathbf{x}) = -\frac{1}{\varepsilon_0}\rho(t,\mathbf{x}) + \frac{1}{4\pi\varepsilon_0 c^2}\int d^3y\,\frac{\ddot{\rho}(t_{\text{ret}},\mathbf{y})}{R}. \tag{1.11}$$

Putting them together, we obtain

$$\Box\phi(t,\mathbf{x}) = -\frac{1}{\varepsilon_0}\rho(t,\mathbf{x}). \tag{1.12}$$

By simple analogy, we have

$$\Box\mathbf{A}(t,\mathbf{x}) = -\mu_0\mathbf{J}(t,\mathbf{x}), \tag{1.13}$$

$$\frac{\partial}{\partial t}\mathbf{A}(t,\mathbf{x}) = \frac{\mu_0}{4\pi}\int d^3y\,\frac{\dot{\mathbf{J}}(t_{\text{ret}},\mathbf{y})}{R}, \tag{1.14}$$

and

$$\nabla A_i(t,\mathbf{x}) = -\frac{\mu_0}{4\pi}\int d^3y\,\hat{\mathbf{R}}\left(\frac{1}{R^2}J_i(t_{\text{ret}},\mathbf{y}) + \frac{1}{cR}\dot{J}_i(t_{\text{ret}},\mathbf{y})\right) \tag{1.15}$$

In the far-field, we have $R \gg 1$ and $|\mathbf{x}| \approx R$, which gives

$$
\begin{aligned}
\mathbf{E}(t,\mathbf{x}) &= -\nabla\phi(t,\mathbf{x}) - \frac{\partial \mathbf{A}(t,\mathbf{x})}{\partial t} \\
&\approx \frac{1}{4\pi\varepsilon_0}\int d^3y\,\hat{\mathbf{x}}\frac{1}{c|\mathbf{x}|}\dot{\rho}(t_{\text{ret}},\mathbf{y}) - \frac{\mu_0}{4\pi}\int d^3y\,\hat{\mathbf{x}}\frac{1}{|\mathbf{x}|}\dot{J}_i(t_{\text{ret}},\mathbf{y}) \\
&= \frac{\mu_0}{4\pi|\mathbf{x}|}\int d^3y\left(\hat{\mathbf{x}}c\dot{\rho}(t_{\text{ret}},\mathbf{y}) - \dot{\mathbf{J}}(t_{\text{ret}},\mathbf{y})\right).
\end{aligned} \tag{1.16}
$$

To simplify the first term, consider the continuity equation,

$$
\dot{\rho}(t_{\mathbf{ret}},\mathbf{y}) + \nabla_y\cdot\mathbf{J}(t,\mathbf{y})\Big|_{t=t_{\text{ret}}} = 0. \tag{1.17}
$$

However, to perform an integration by parts, we need

$$
\nabla_y\cdot\mathbf{J}(t_{\text{ret}},\mathbf{y}) = \nabla_y\cdot\mathbf{J}(t,\mathbf{y})\Big|_{t=t_{\text{ret}}} + \dot{\mathbf{J}}(t_{\text{ret}},\mathbf{y})\cdot\nabla t_{\text{ret}} = \nabla_y\cdot\mathbf{J}(t,\mathbf{y})\Big|_{t=t_{\text{ret}}} + \dot{\mathbf{J}}(t_{\text{ret}},\mathbf{y})\cdot\frac{\hat{\mathbf{R}}}{c}. \tag{1.18}
$$

Thus, the first term integral becomes

$$
\frac{\mu_0}{4\pi|\mathbf{x}|}\int d^3y\,\hat{\mathbf{x}}\dot{\mathbf{J}}(t_{\text{ret}},\mathbf{y})\cdot\hat{\mathbf{R}} - \frac{\mu_0}{4\pi|\mathbf{x}|}\oint d^2y\,\mathbf{J}(t_{\text{ret}},\mathbf{y})\cdot\hat{\mathbf{n}}. \tag{1.19}
$$

Since $\mathbf{J}$ is bounded, the surface integral vanishes. Therefore, we can write

$$
E(t,\mathbf{x}) \approx \frac{\mu_0}{4\pi|\mathbf{x}|}\int d^3y\left(\hat{\mathbf{x}}(\dot{\mathbf{J}}(t_{\text{ret}},\mathbf{y})\cdot\hat{\mathbf{R}}) - \dot{\mathbf{J}}(t_{\text{ret}},\mathbf{y})\right), \tag{1.20}
$$

Again, for far-field, we have $\hat{\mathbf{R}} \approx \hat{\mathbf{x}}$. Finally, using the vector identity, we find

$$
\mathbf{E}(t,\mathbf{x}) \approx \frac{\mu_0}{4\pi|\mathbf{x}|}\,\hat{\mathbf{x}}\times\left(\hat{\mathbf{x}}\times\int d^3y\,\dot{\mathbf{J}}(t_{\text{ret}},\mathbf{y})\right). \tag{1.21}
$$

For the magnetic field, we have

$$
B_i(t,\mathbf{x}) = \epsilon_{ijk}\partial_j A_k \approx -\frac{\mu_0}{4\pi c|\mathbf{x}|}\epsilon_{ijk}\int d^3y\,\hat{\mathbf{x}}_j\dot{J}_k(t,\mathbf{x}), \tag{1.22}
$$

where we have made a far-field approximation again. Therefore, we find

$$
\mathbf{B}(t,\mathbf{x}) \approx -\frac{\mu_0}{4\pi c|\mathbf{x}|}\,\hat{\mathbf{x}}\times\int d^3y\,\dot{\mathbf{J}}(t_{\text{ret}},\mathbf{y}) = \frac{1}{c}\hat{\mathbf{x}}\times\mathbf{E}(t,\mathbf{x}). \tag{1.23}
$$

The Poynting vector of the radiation is, therefore, given by

$$
\mathbf{S(t,x)} = \frac{1}{\mu_0 c}\mathbf{E}(t,\mathbf{x})\times\left(\hat{\mathbf{x}}\times\mathbf{E}(t,\mathbf{x})\right) = \frac{1}{\mu_0 c}[\hat{\mathbf{x}}|\mathbf{E}(t,\mathbf{x})|^2 - \mathbf{E}(t,\mathbf{x})(\hat{\mathbf{x}}\cdot\mathbf{E}(t,\mathbf{x}))]. \tag{1.24}
$$

Note that $\hat{\mathbf{x}}\cdot\mathbf{E} = 0$, and hence we have

$$
\mathbf{S(t,x)} = \frac{\mu_0}{16\pi^2 c|\hat{\mathbf{x}}|^2}\left|\int d^3y\dot{\mathbf{J}}(t_{\text{ret}},\mathbf{y})\right|^2\sin^2\theta\,\hat{\mathbf{x}}, \tag{1.25}
$$

where θ is the angle between the integral and $\hat{\mathbf{x}}$.